

to allocate once more to Excel. This is usually associated with thrashing of the hard drive: your system steadily blinks a red light to indicate constant disk access.

Scenarios such as this are usually due to inefficient memory management. Vista aims to eradicate, or at least alleviate these issues to a large degree by implementing memory priorities, just as it does with I/O priorities.

Memory is cached by the system in a buffer called the Standby List in the form of “pages.” When a program requests memory from the system, the system creates a new page at the beginning of the Standby List with the help of the kernel’s Memory Manager. When the program is done using the memory space, its allocated page is cached to the end of the Standby List—if the program wants to access the page again, the Memory Manager is called to reassign the page to the program.

Under Vista, each Standby List is further divided into eight parts based on priorities: thus every page of memory has a priority in the range of 0 to 7. The page inherits the priority from the program or thread that creates the page. Such a Standby List which is arranged by priority is then used by the Memory Manager to better anticipate future memory calls of an application and to thus make allowances for each application based on its historical usage. To better understand how this happens, we need to take a closer look at how SuperFetch works.

SuperFetch is a memory management scheme which is used to predict memory requirements of an application based on its historical usage of such resources. SuperFetch makes application launches quicker as time passes. It does so by tracking which applications you often launch, and at what time, etc. and then using this historical information to preload parts of the frequently-used programs into memory, thus ensuring that future sessions of the application start faster.